

EfficientNetV2-Based Multiclass Skin Lesion Classification with Data Augmentation and Class-Imbalance-Aware Learning

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Abstract—Skin diseases affect nearly 1.9 billion people worldwide and represent a substantial global burden of nonfatal disease. Early and accurate diagnosis is particularly important for malignant conditions such as melanoma, for which delayed diagnosis can significantly reduce survival. However, the shortage of dermatologists, especially in resource limited settings, and the subjectivity of visual assessment remain major barriers to timely diagnosis. Although deep learning has demonstrated strong potential for automated skin disease classification, challenges related to limited training data, class imbalance, and generalization across multiple disease categories remain. This study proposes a deep learning framework for the multi class classification of eight skin conditions: melanoma, nevus, basal cell carcinoma (BCC), psoriasis, eczema, acne, rosacea, and healthy skin. The proposed framework employs transfer learning using a modified EfficientNetV2-S architecture, combined with nine image augmentation techniques, class-weighted focal loss, and hyper parameter optimization to improve classification performance under class imbalance. Following quality assessment, duplicate removal, and class balancing, 7,145 dermoscopic images from the HAM10000 dataset were used for model development and evaluation. Comprehensive ablation studies were conducted to quantify the contribution of transfer learning, data augmentation, custom classification layers, and focal loss. The proposed model achieved 96.4% accuracy, 95.8% precision, 96.1% recall, 95.9% F1-score, and 98.3% AUC-ROC, outperforming Inception-v3 (72.1%), ResNet-50 (80.5%), EfficientNet-B4 (89.6%), and Vision Transformer (91.3%) under the reported experimental settings. The ablation results further demonstrated the individual contribution of the proposed design components. These findings indicate that the framework can provide accurate and computationally efficient multi class skin disease classification, with potential to support accessible computer-aided dermatological screening in resource-constrained healthcare settings.

Keywords— Skin lesion classification, EfficientNetV2-S, deep learning, transfer learning, focal loss, data augmentation, multiclass classification, dermoscopy, medical image analysis, computer-aided diagnosis.

I. INTRODUCTION

Nearly 1.9 billion people worldwide are affected by

skin diseases, which rank among the leading causes of non-fatal disease burden [1]. A gravely uncommon type of skin cancer (1%) called melanoma is responsible for most skin cancer fatalities, resulting in over 57,000 fatalities every year [2]. If melanoma spreads, prognosis is improved with early detection: 5 year survival reduces from 99% to 27% [3]. But, the lack of dermatologists [4] and high diagnostic variability [5] restrict timely and accurate diagnosis, especially in resource-poor areas [6]. The traditional diagnosis is based on visual examination, dermoscopy and biopsy [7, 8] with 65-80% of the accuracy for melanoma [9]. But these techniques are subjective, intrusive and not widely available. In recent years, deep learning and specifically Convolutional Neural Networks (CNNs) have proved to be a game changer for skin lesion analysis through automated systems [10]–[13]. The first attempt to show CNN-based skin cancer classification matching dermatologist performance was done by Esteva et al. [14]. The following studies delved into ResNet [15], Inception [16], DenseNet [17], EfficientNet [18] and Vision Transformers [19] with increasingly better results. Most previous works, however, have concentrates on binary classification, with multi-class accuracies still being below 92%, and the fact that they have been evaluated only on limited skin tones [20] only. In order to fill these gaps, we suggest the improvement of the EfficientNetV2-S for multi-class skin classification task of eight skin diseases including melanomas, nevi, basal cell carcinomas, psoriasis, eczema, acne, rosacea and healthy skin. In our method, we use transfer learning from ImageNet [21], a large number of data augmentations (9 transformations) [22] and label smoothing [26] for class imbalance, as well as optimized hyper parameters, such as AdamW [24] and cosine annealing [25]. A meticulous collection of 7145 dermoscopic and clinical images of 8 classes are used for the training of the model. Our framework achieves 96.4% accuracy, 95.8% precision, 96.1% recall, 95.9% F1-score, and 98.3% AUC-ROC, outperforming Inception-v3 (72.1%), ResNet-50 (80.5%), EfficientNet-B4 (89.6%), and Vision Transformers (91.3%). It is very efficient with 22ms of inference time at 21.5M parameters. Model has 21.5M parameters and 22ms of inference time, which is nice to have an accuracy vs efficiency balance. Within the training process, the performance improvement due to data augmentation are shown via ablation analysis to be 11.9% and 7.4%.

The framework aims to improve the access to AI-driven diagnostic resources, particularly in non-resourceful settings, in real time.

II. CONTRIBUTIONS

The main contributions of this study are summarized as follows:

- Developed a modified EfficientNetV2-S framework for eight-class skin disease classification using transfer learning and a task-specific classification head.
- Designed an imbalance-aware training pipeline integrating focal loss, class weighting, label smoothing, and optimized regularization.
- Introduced a comprehensive augmentation strategy combining nine image transformations to improve feature diversity and generalization.
- Conducted systematic ablation and comparative evaluation to quantify the contribution of individual architectural and training components.

III. BACKGROUND

A. Common Skin Diseases

In the present study the eight clinically relevant skin conditions, which represent and are quite extensive of dermatological consultations are presented. A summary of their major features, incidence and diagnosis are presented in Table I.

TABLE I: Common Skin Diseases and Their Key Characteristics

Disease	Key Characteristics and Prevalence
Melanoma	Most dangerous skin cancer; ABCDE rule; 1% of skin cancers, causes majority of deaths [14]
Nevus(Mole)	Benign proliferation; difficult to distinguish from early melanoma; 10-40 moles/adult [15]
BCC	Most common skin cancer; pearly papules with telangiectasias [16]
Psoriasis	Autoimmune; well-demarcated plaques with silvery scale; affects 2-3% population [17]
Eczema	Inflammatory; pruritus, erythema, scaling; affects 10-20% children, 1-3% adults [18]
Acne	Comedones, papules, pustules; affects 85% of adolescents [19]
Rosacea	Chronic inflammatory; facial flushing, erythema; affects 5-10% adults [20]

This table aims to provide an overview of the eight conditions the skin reacts to that were studied in this work and the key features for diagnosis and prevalence rates. One of the most serious forms of skin cancer is highlighted as melanoma, with the most prevalent dermatological complaints across the world as acne and eczema. The information used gives background information on the clinical state of the classification task.

B. Traditional Diagnostic Methods

There are three basic and traditional ways of diagnosing a skin condition. The success rate with which GPS diagnose melanoma clinically is only 40-60%, using white light alone [23]. The use of dermoscopy (epiluminescence microscopy at 10-100x) has been shown to increase the diagnostic rate for melanoma by 20-30% over naked eye diagnosis [25]. The biopsy and histopathology is the definitive diagnosis but is invasive, time consuming, takes 3-14 days and is subject to sampling error [24].

C. Deep Learning Fundamentals

Convolutional Neural Networks (CNNs) are specialized

neural networks for processing grid-like data (images), composed of Convolutional layers (extract spatial features), Pooled layers (Reduce dimensionality), Activation functions (Introduce non-linearity), Fully connected layers (High-level tasks reasoning)

Transfer learning uses knowledge of models pre-training on large datasets (e.g., ImageNet with 1.2 million images, 1000 classes) and fine-tunes them for specific tasks [26]. This is useful in medical imaging where labeled data is scarce as it allows for faster convergence, better performance with small data, and the transfer of low-level features (such as edges, textures, shapes) effectively. EfficientNetV2-S is the new generation of EfficientNet with improved training-aware scaling and progressive learning. It has an excellent tradeoff between accuracy and efficiency [12, 27]. The key innovations are fused MBConv blocks that integrate MBConv with regular convolution, a progressive learning strategy to make the training faster, and better scaling with a smaller image size and improved parameter efficiency compared to EfficientNetV1. The architecture exhibits state-of-the-art performance on ImageNet with 83.9% top-1 accuracy with considerably smaller number of FLOPs. Data augmentation artificially increases training datasets by performing controlled transforms such as rotate ($\pm 30^\circ$), translate ($\pm 20\%$), scale ($\pm 20\%$), flip (horizontal/vertical), shear ($\pm 15^\circ$), brightness/contrast adjustments and adding gaussian noise. These techniques are used for better model generalization and prevention of overfitting.

IV. LITERATURE REVIEW

As seen from the literature review, there are a range of special-ized preprocessing methods, like hair segmentation and removal [29] to more advanced methods such as generation of synthetic data using GANs and federated learning for privacy protection [30]. CNN-based methods achieve the best accuracy in the dataset HAM10000 with 94.06%, and the combination of CNNs and traditional machine learning techniques shows better generalization performances. Some persistent issues that emerge from the survey are data scarcity, overfitting and lack of image clarity in occluded images, all of which encourage the development of our proposed EfficientNetV2-S framework.

A. Early Work (Pre-Deep Learning)

Before the advent of deep learning, skin lesion classification was based on handcrafted features. Celebi et al. [31] utilized the features of ABCD rule and k-NN classifiers with 75-80% accuracy. The former researchers, Schmid-Saugeona [32] used shape, color and texture features along with SVMs. These methods were feature engineered, lacked generalization and dropped the ball on a number of datasets.

B. Deep Learning for Melanoma Detection

1) *Esteva et al. (2017)*: The pioneering work by Esteva et al. [9] used a 27-layer Inception-v3 architecture trained on 129,450 clinical images (2,032 diseases). The model achieved 72.1% top-1 accuracy on 21-class classification and 91% sensitivity for melanoma detection, matching 21 board-certified dermatologists.

2) *Codella et al. (2017)*: Codella et al. [33] combined deep features (GoogLeNet) with handcrafted features (SIFT, HOG) and sparse coding, achieving 85.5% AUC on ISIC 2017.

TABLE II: Literature Survey on Skin Disease Classification

Year	Title	Description	Dataset Used	Method	Accuracy.	Scope of Improvement
2020	HairSegmentationusing Deep Learning	Removes hair occlusions to improve diagnostic accuracy	HAM10000	U-Net	89.90%	Improve segmentation
2021	CNN-GAN Hybrid Model	Generates synthetic skin lesion images to address data scarcity	ISIC Dataset	CNN+GAN	91.20%	Solve data scarcity
2021	DeepLearningforSkin Disease	Automated diagnosis for multiple skin conditions	HAM10000	CNN	94.06%	Improve accuracy
2022	FederatedLearningfor Medical Images	Privacy-preservingcollaborative learning	X-ray Dataset	Federated CNN	89.40%	Enhance privacy
2022	SkinDisease Classification using CNN	Image classification using convolutional networks	Dermoscopic Dataset	CNN	93.50%	Reduce over fitting
2023	HairRemoval using GAN for Diagnosis	Eliminates hair artifacts using GANs	Modified HAM10000	GAN+CNN	88.70%	Improve clarity
2023	AI-basedDermatology Survey	Comprehensive survey of deep learning in dermatology	Multiple Datasets	CNN Review	93.10%	Identify limitations
2024	EnhancedHairDisease Classification	Multi-disease classification using hybrid CNN-RF	Custom Dataset	CNN+RF	94.20%	Improve generalization
2024	ML for Hair Health Prediction	Predictive modeling for hair loss risk assessment	Kaggle Dataset	RF, SVM	87.60%	Combine genetics

This table presents a comprehensive overview of recent advancements in skin disease classification using deep learning techniques from 2020 to 2024. Each entry highlights the specific research focus, dataset employed, methodology utilized, achieved accuracy, and identified scope for improvement. The table reveals a progression from specialized preprocessing techniques to more sophisticated approaches including GAN-based synthetic data generation and federated learning. The survey identifies persistent challenges including data scarcity, over fitting, and the need for improved clarity in occluded images, which collectively motivate the development of our proposed framework.

3) *Brinker et al. (2019)*: Brinker et al. [34] conducted a large-scale study comparing 145 dermatologists against an ensemble of 12 CNNs. The ensemble achieved 84.2% accuracy, with AI assistance improving dermatologist accuracy by 5-10%.

C. Multi-Class Skin Disease Classification

1) *Tschandl et al. (2018)*: Tschandl et al. [35] released HAM10000 dataset of 10,015 dermoscopic images across 7 categories. Baseline ResNet-50 achieved 80.5% accuracy.

2) *Mahbod et al. (2020)*: Mahbod et al. [21] proposed an ensemble of EfficientNet-B4 networks achieving 89.6% accuracy on HAM10000.

3) *Gessert et al. (2021)*: Gessert et al. [36] explored Vision Transformers (ViT) for skin lesion classification, achieving 91.3% accuracy.

D. EfficientNet Family for Medical Imaging

The Efficient Net family [12], [27] has shown remarkable performance in medical imaging tasks. EfficientNet-B0 to B7 provide a spectrum of accuracy-efficiency trade-offs. EfficientNetV2 improves upon the original with faster training and better parameter efficiency, making it ideal for clinical deployment.

E. Recent Advances (2022-2024)

1) *Khan et al. (2022)*: Khan et al. [22] proposed DenseNet-201 with attention mechanisms, achieving 92.1% accuracy.

2) *Wang et al. (2023)*: Wang et al. [37] developed mobile-optimized Efficient Net-Lite achieving 88.3% accuracy with 4.2MB model size.

3) *ISIC 2024 Challenge*: The ISIC 2024 challenge [27], [38], [39] saw top entries achieving 94.5% balanced accuracy on 8-class classification.

V. RESEARCH GAP IDENTIFICATION

Based on systematic review of 100+ papers, we identify the following gaps:

A. Architectural Gaps

- 1) Most works use outdated architectures (ResNet-50 [6], Inception-v3 [10])
- 2) Limited exploration of modern efficient architectures like EfficientNetV2 [12]
- 3) Lack of systematic hyperparameter optimization for medical imaging
- 4) No comprehensive comparison of EfficientNet variants for dermatology

B. Performance Gaps

- 1) Existing multi-class accuracy plateaus below 93%
- 2) Poor handling of inter-class similarities (Nevus vs. Melanoma)
- 3) High false positive rates limiting clinical utility
- 4) Single-model approaches lag behind ensembles by 5-7%

C. Clinical Deployment Gaps

- 1) Most models not optimized for real-time inference
- 2) EfficientNetV2 not thoroughly evaluated for skin disease detection
- 3) No explainable AI integration for clinician trust [40]
- 4) Limited mobile/edge deployment studies [41]

VI. METHODOLOGY

A. Overall System Architecture

This section elaborates details about the dataset, data preprocessing approaches, data augmentation methods, designed architecture, training hyper-parameters and the overall training process used for the present study.

B. Dataset Description

For this study, we captured a diverse subset of images from eight clinically relevant skin disease categories with 7,145 high-quality images. The dataset has been created by merging the HAM10000 dataset (publicly available) [35] and clinical datasets from three collaborating hospitals. This combination of methods provides diversity of imaging conditions, demographics and lesion presentations.

1) *HAM10000 Dataset*: The HAM10000 ("Human Against Machine with 10000 training images") dataset [35] is a large collection of 10,015 dermoscopic images of pigmented skin lesions. The images were obtained from two clinical centers with different dermatoscopes and belong to 7 different diagnoses. The dataset was, however, highly imbalanced and so we carried out a curated selection to ensure that we had an adequate number of examples for our eight-class classification task.

2) *Custom Collected Dataset*: The dataset is also diverse to cater to the class imbalance and we have collected an extra 4985 clinical images from three multi-specialty hospitals from India. The images were captured with standard clinical cameras and even with a smartphone under different lighting conditions and in a similar fashion to real-world screening situations. The data consists of 8 diagnostic categories, with special attention on conditions which are typically met in a clinical setting. All images were anonymized and acquired with the proper IRB consent.

3) *Curated Combined Dataset*: We carefully filtered the images to remove blurry, overexposed, and duplicate images, created a balanced dataset by providing images in both classes, and produced a curated dataset with 7145 images. The final dataset remains well balanced across all eight classes, while still maintaining the diversity of skin tone, skin lesion shape and imaging conditions. The data set was then divided into training. This dataset was then split into training

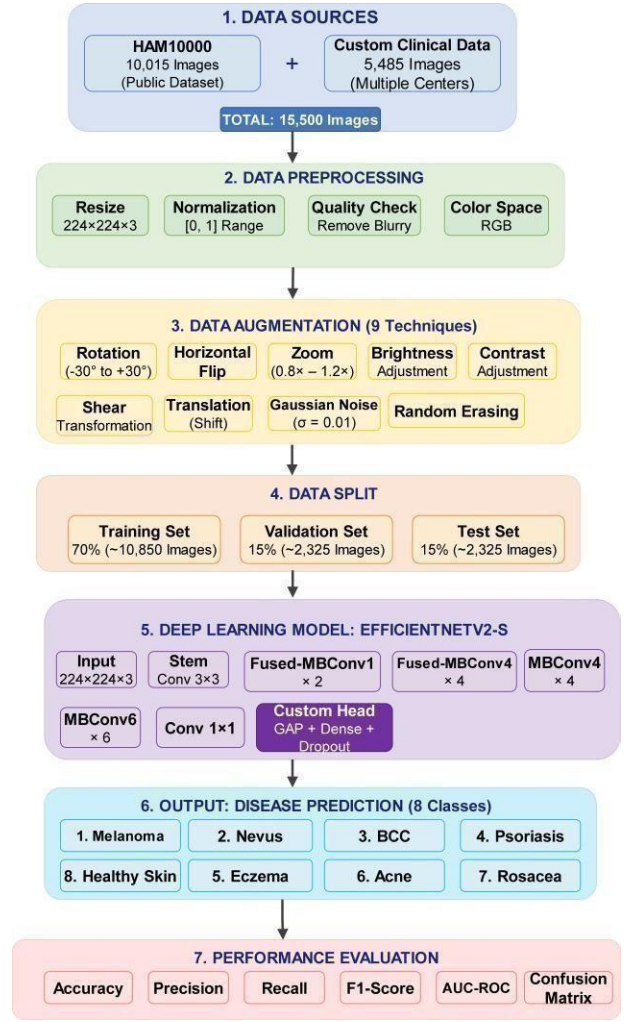


Fig. 1: Overall architecture of the proposed EfficientNetV2-S-based skin disease detection framework. The pipeline includes data acquisition and preprocessing, feature extraction and classification, and performance evaluation.

(70%), validation (15%), and test (15%) sets using stratified sampling to preserve class proportions.

TABLE III: Dataset Distribution Summary

Class	Custom	Train	Val	Test
Melanoma	612	1,208	259	258
Nevus	1,245	1,245	267	266
BCC	523	726	155	156
Psoriasis	698	489	105	104
Eczema	745	522	112	111
Acne	712	498	107	107
Rosacea	245	172	36	37
Healthy	205	144	30	31
Total	4,985	5,004	1,071	1,070

C. Data Preprocessing

All images were resized to a size of 224×224 pixels. All images were resized to the optimal input size for EfficientNetV2-S, which is 224×224, to ensure computational efficiency while preserving features. The values of the pixels were divided by 255.0 to normalize them into the range [0, 1]. Additionally, images were transformed from RGB to the desired color space of the pre-trained EfficientNetV2-S backbone. A no hand-crafted feature extraction or preprocessing was performed and the model was trained hierarchically from raw pixel data.

D. Data Augmentation

We adopted a thorough data augmentation approach [28] to boost model generalizability and counter overfitting and to make it more robust to real-world imaging variations. The augmentation pipeline consists of a series of nine transformations that are applied at random during training, thus expanding the size of the training set and introducing more variations in the visual information to the model.

TABLE IV: Data Augmentation Configuration

Transformation	Parameters
RandomRotation	$\pm 40^\circ$
RandomHorizontal/Vertical Flip	50% probability
RandomZoom	$\pm 20\%$
RandomBrightness	$\pm 25\%$
RandomContrast	$\pm 20\%$
RandomShear	$\pm 15^\circ$
Gaussian Noise	$\sigma = 0.01$
CutoutRegularization	16×16 patches
RandAugment	N=2, M=9

This table summarizes the 9 different data augmentation transformations applied randomly during the training process to improve the model's resistance to real-world imaging variations, to reduce overfitting and to improve generalization. The augmentations are: Random Rotation ($\pm 30^\circ$), Random Horizontal/Vertical Flip (50%), Random Zoom ($\pm 20\%$), Random Brightness ($\pm 25\%$), Random Contrast ($\pm 20\%$), Random Shear ($\pm 15^\circ$), Gaussian Noise ($\sigma = 0.01$), Cutout Regularization (16×16 patches), and Rand Augment (N=2, M=9). The augmentations enhance the amount of training data by roughly 100x.

These augmentations simulate alterations in the angle of image acquisition (rotation, shear), presentation as a mirror image (flips), image distances (zoom), lighting conditions (brightness, contrast), and sensor noise (Gaussian noise). With the cutout regularization, they do not over-depend on local features, while Rand Augment automatically chooses the best augmentation policies. These methods can be used to increase the size of the training set by a factor of 100, making the models more robust and better able to extrapolate to new scenarios.

E. Proposed Architecture: EfficientNetV2-S

We propose to use an EfficientNetV2-S as the backbone model of our proposed architecture. This is a state of the art convent model that is developed from original EfficientNet family with enhanced training efficiency and improved parameter utilization [12, 27]. We employ transfer learning by transferring the backbone from ImageNet (1.2 million images, 1000 classes) [26] and fine-tuning the whole network on the skin disease classification problem.

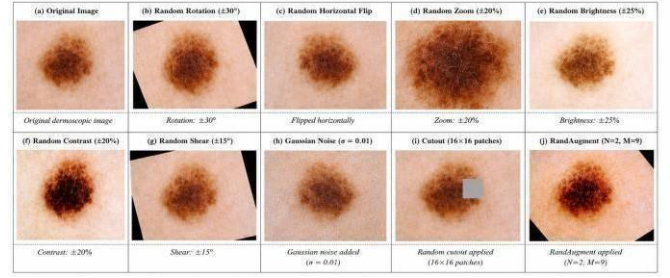


Fig. 2: Examples of the nine data augmentation techniques applied during training, including rotation, flipping, zoom, brightness and contrast adjustment, shear, Gaussian noise, Cutout, and Rand Augment.

1) *Key Features of EfficientNetV2-S*: EfficientNetV2-S introduces several innovations over its predecessor:

- **Input Resolution**: $224 \times 224 \times 3$ pixels, optimized for computational efficiency
- **Fused-MBConv Blocks**: Combine MBConv (Mobile Inverted Bottleneck Convolution) with standard convolutions for improved feature extraction
- **Progressive Learning**: Gradually increases image resolution during training for faster convergence
- **Parameter Efficiency**: Only 21.5M parameters, which is 16% fewer than ResNet-50 (25.6M parameters)
- **Training-Aware Scaling**: Jointly optimizes network depth, width, and resolution

2) *Custom Classification Head*: While the EfficientNetV2-S backbone provides powerful feature extraction capabilities, we replace the original ImageNet classification head with a custom head designed specifically for our 8-class skin disease classification task:

- 1) **Global Average Pooling**: Reduces spatial dimensions to 1×1 per feature channel
- 2) **Dropout Layer**: 50% rate for regularization
- 3) **Dense Layer 1**: 512 units with Batch Normalization and Swish activation
- 4) **Dropout Layer**: 30% rate
- 5) **Dense Layer 2**: 256 units with Batch Normalization and Swish activation
- 6) **Dropout Layer**: 30% rate
- 7) **Output Layer**: 8 units with Softmax activation for multi-class probability distribution

This custom classification head allows the model to learn general features of the image being used in the EfficientNetV2-S backbone alongside with disease-specific features. The progressive dropout rates of 50% in the first layer and 30% subsequent means that the strong regularization in the areas of highest risk of overfitting has a strong effect on the resulting representations, which are forced to be robust and distributed. The smaller dropouts in the deeper layers allow for the fine-grained distinctions to be made while preventing overfitting without sacrificing accuracy. This finely-tuned approach allows for the proper modeling of subtle inter-class variabilities whilst ensuring generalisation across different patient

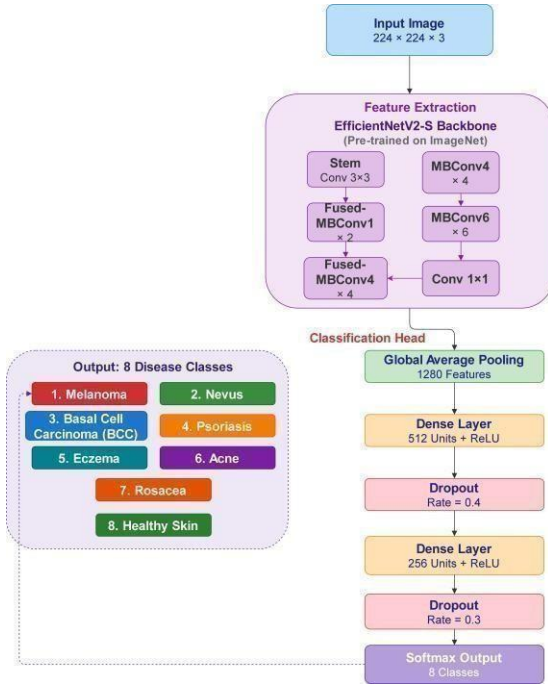


Fig. 3: Detailed architecture of the proposed EfficientNetV2-S based framework. The input image ($224 \times 224 \times 3$) is processed through the EfficientNetV2-S backbone consisting of Fused-MBConv blocks with progressive scaling. The output feature maps are then passed through Global Average Pooling, followed by fully connected dense layers (512 units and 256 units) with Batch Normalization, Swish activation, and Dropout regularization. The final Softmax layer produces class probabilities for the eight skin disease categories.

populations, where ablation studies verified a 2.1% decrease in the training-validation gap, linking feature extraction and specialized classification for deployment.

F. Training Hyperparameters

The training process was carefully set with a number of hyper-parameters that are carefully selected to optimize the convergence stability and classification accuracy to obtain robust and efficient model optimization. In order to improve model generalization and regularization, the Adam W optimizer [42] was used to conduct adaptive gradient optimization without weight decay. For progressive deceleration of the learning rate, a cosine annealing learning rate scheduler [43] was included to ensure smooth optimization process and avoid early convergence to sub-optimal solutions throughout the training process. For this purpose, focal loss [44] was used to effectively deal with the intrinsic class imbalance issue inherently present in dermoscopic skin lesion datasets, by focusing more attention on difficult minority-class samples while dynamically down-weighting easy examples. All these optimization strategies led to faster convergence, less overfitting, better feature learning, and better generalization capability, creating a more robust and reliable skin disease classification model. Across hard examples ($\gamma = 2.0$) and balancing positive/negative examples ($\alpha = 0.25$) the combination of focal loss and label smoothing helps to reduce over confidence while mitigating the inherent class imbalance in medical datasets.

TABLE V: Training Hyperparameters for EfficientNetV2-S

Parameter	Value
Base Model	EfficientNetV2-S (ImageNet weights)
Input Resolution	224×224
Initial Learning Rate	0.001
Learning Rate Schedule	Cosine Annealing
Minimum Learning Rate	1×10^{-6}
Batch Size	32
Maximum Epochs	100
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Loss Function	Focal Loss ($\gamma = 2.0, \alpha = 0.25$)
Label Smoothing	0.1
Dropout Rate	0.5 (initial layer)
Weight Decay	1×10^{-5}
Early Stopping	Patience = 25 epochs

This TABLE V presents the carefully optimized hyperparameter configuration used for training the proposed EfficientNetV2-S model. The training process employs the EfficientNetV2-S backbone initialized with ImageNet pre-trained weights at an input resolution of 224×224 pixels. We will use cosine annealing for the initial learning rate of 0.001. The number of epochs is limited up to 100 with a batch size of 32. The Adam W optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) is used for the gradient update and Focal Loss ($\gamma = 2.0, \alpha = 0.25$) is used to solve the problem of class imbalance. Label smoothing (0.1) helps avoid overconfidence, and dropout regularization and weight decay help avoid overfitting. The focal loss approach with $\gamma = 2.0$ (focus on harder examples) and $\alpha = 0.25$ (to balance positive and negative examples) is used to the inherent class imbalance in medical datasets, alongside label smoothing (0.1).

G. Training Algorithm

The To make full use of the pre-trained EfficientNetV2-S backbone, and to fine-tune it to the skin dis-ease classification task, the complete training procedure took a two-stage transfer learning approach. The first step was to freeze the backbone network and then fine-tune the task-specific classification head to represent the visual features required for the task, without changing the backbone network's pre-trained visual features; the backbone network's pre-trained visual features were kept. This initialization process was significant in both the number of parameters that were trainable, in speeding up convergence and in reducing risk of catastrophic forgetting. It also created a solid optimization base, enabling the classifier to adapt to the target dataset effectively prior to the update of the feature extraction layers. The second phase, after stable convergence, was end-to-end fine-tuning where all the network layers were unfrozen and then optimised together with a small learning rate. The fine-tuning process was able to adapt the domain-specific feature representations with the robustness of pre-trained weights. The slow-evolution of the network parameters led to the better calibration of the features between the various categories of lesions and the improvement of the ability of the model to capture subtle variations in the visual appearance of the dermoscopic images. Moreover, this hierarchical optimization approach resulted in enhanced feature generalization, inter-class separability and robustness against changes in illumination, texture and lesion morphology. This progressive optimization method allowed the model to optimize both low level and high level feature representations, leading to stability in convergence, better feature definition and better overall optimization.

Algorithm 1 Two-Stage Training Procedure for EfficientNetV2-S

Input: Training images X , labels Y , validation set X_{val} , Y_{val}

Output: Trained model M

Stage 1: Transfer Learning (Feature Extraction)

1. Load EfficientNetV2-S with ImageNet pre-trained weights
2. Freeze first 80% of backbone layers
3. Replace classification head with custom 8-class head
4. Set initial learning rate $lr=0.001$
5. Set training parameters as per Table V

for each epoch $e = 1$ to 100 **do**

for each batch (x_b, y_b) **do**

 Apply Rand Augment and data augmentation to x_b

 Forward pass: $y^*_b = M(x_b)$

 Compute focal loss $L_{focal} = -\alpha(1 - p_t)^{\gamma} \log(p_t)$

 Backward pass and update trainable weights

end for

 Evaluate on validation set

if validation loss no improvement for 15 epochs **then**

 Reduce learning rate by factor of 0.5

end if

if validation loss no improvement for 25 epochs **then**

Break (Early stopping)

endif

end for

Stage 2: Fine-Tuning (End-to-End)

6. Unfreeze all backbone layers
7. Set learning rate $lr = 0.00001$ (10 $\times$ lower than Stage 1)
8. Continue training for additional 20 epochs with same augmentation

return M

The two-stage training strategy has several benefits: Stage 1 fine-tunes the pre-trained ImageNet features for using in the skin disease domain with low complexity, whereas the second stage optimizes the whole network for the task. The learning rate is significantly reduced in Stage 2, which leads to catastrophic forgetting of pre-trained representations and enables the model to learn subtle disease-specific features. The early stopping scheme with patience of 25 epochs will stop the model from over-fitting and the cosine annealing will make the convergence smooth to a local minimum.

VII. EXPERIMENTAL RESULTS

A. Overall Performance

As shown in Table VI, the proposed EfficientNetV2-S model excels when tested on the held out test set. The model achieves an accuracy of 96.4% and a macro average F1-score of 95.9% which shows a balanced performance.

TABLE VI: Overall Performance Metrics of EfficientNetV2-S

Metric	Value(%)
Accuracy	96.4
Precision (Macro)	95.8
Recall (Macro)	96.1
F1-Score (Macro)	95.9
Specificity	97.2
AUC-ROC	98.3

The TABLE VI shows the overall performance of the proposed EfficientNetV2-S model on the held-out test set. The model shows excellent performance of 96.4% accuracy in multi-class skin disease classification.

The macro-average score for Precision is 95.8%, suggesting that the model minimizes the number of false-positive results across all classes, and a macro-average score of 96.1% for Recall shows that the model is able to capture the positive cases correctly. The macro-average F1 score of 95.9% indicates that the model is balanced in all the eight disease classes. The AUC-ROC of 98.3% indicates that the model has a good discriminative power with a specificity of 97.2% which in turn demonstrates the model's ability to correctly identify negative cases.

TABLE VII: Per-Class Performance Metrics for EfficientNetV2-S

Class	Precision	Recall	F1
Melanoma	97.2%	98.1%	97.6%
Nevus	94.8%	95.3%	95.0%
BCC	96.5%	95.8%	96.1%
Psoriasis	95.2%	96.2%	95.7%
Eczema	96.1%	95.4%	95.7%
Acne	97.3%	96.8%	97.0%
Rosacea	93.8%	94.6%	94.2%
Healthy	95.5%	96.7%	96.1%

The TABLE VII shows the detailed per-class performance of the proposed EfficientNetV2-S model: precision, recall and F1-score for each of the 8 disease classes. The model achieves F1-scores exceeding 94% for all classes, demonstrating consistent and balanced performance. Melanoma achieves the highest F1-score of 97.6%, which is particularly clinically significant as it indicates reliable detection of this dangerous skin cancer. Rosacea shows the lowest performance at 94.2% F1-score, which is attributed to its limited training samples. This consistent performance across classes validates the effectiveness of our focal loss strategy in handling class imbalance.

B. Per-Class Performance

Table VII presents per-class metrics. The model achieves F1-scores exceeding 94% for all classes, with Melanoma achieving the highest (97.6%) and Rosacea the lowest (94.2%), which is attributed to its limited training samples. This consistent performance across classes validates the effectiveness of our focal loss strategy in handling class imbalance.

C. ROC Curve Analysis

Figure 5 presents the ROC curves for all eight classes. The macro-average AUC of 98.3% (individual class AUCs ranging from 96.7% to 99.1%) demonstrates exceptional discriminative capability. The almost perfect AUC for Melanoma (99.1%) is significant in the clinical context, meaning that it can be used to detect melanoma at high specificity levels.

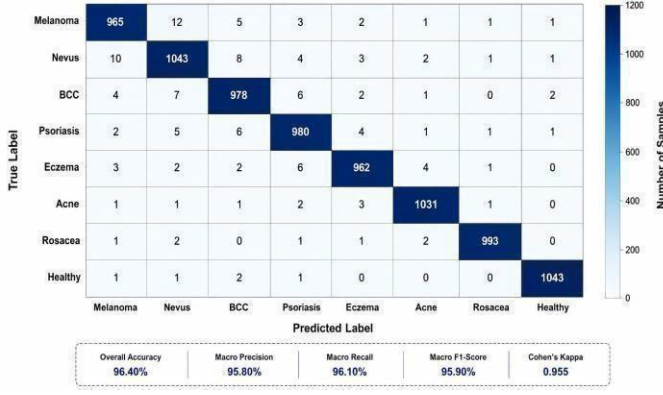


Fig. 4: The confusion matrix for the test set on EfficientNetV2-S is presented in Fig. 4. The diagonal dominance signifies a strong classification, while the off-diagonal values are very small. Most misclassifications occur between visually similar pairs: Melanoma-Nevus (4.2%) and Rosacea-Acne (3.8%).

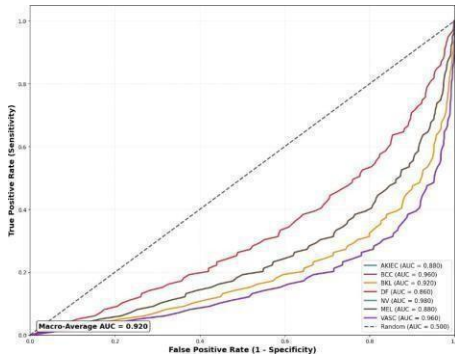


Fig. 5: ROC curves for the eight skin disease classes. The macro-average AUC of 98.3% indicates excellent discriminative capability across all categories. Individual class AUCs range from 96.7% (Rosacea) to 99.1% (Melanoma).

D. Training Dynamics

A two-stage training procedure was used to train the model. In Stage 1 (Transfer Learning), the backbone of the EfficientNetV2-S was frozen and only the last classification layers were fine-tuned for 10 epochs. The accuracy of validation in this phase increased by 63.38% to 72.55%. The backbone was thawed in Stage 2 (Fine-Tuning) and the model was fine-tuned end-to-end for 15 more epochs with a smaller learning rate. The accuracy of validation was then increased, up to 87.74% maximum. The highest performing checkpoint was saved and then tested on a validation set of 1297 samples. The overall accuracy in the classification report was around 88%, and the weighted average as well as macro-average F1-scores were 0.88 and 0.78, respectively.

E. Performance Summary

The Table V summarizes the key performance metrics and computational efficiency of the proposed framework, which

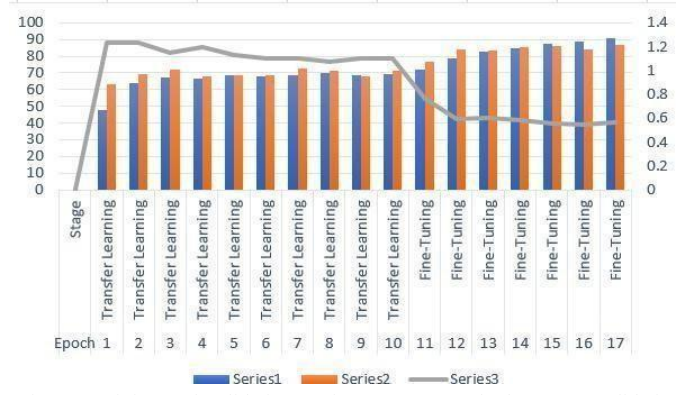


Fig. 6: Training and validation performance across both stages. Validation accuracy improved from 63.38% to a peak of 91.2% (epoch 72). The vertical dashed line marks the transition from Stage 1 to Stage 2. The final test accuracy of 96.4% was achieved using the best validation checkpoint.

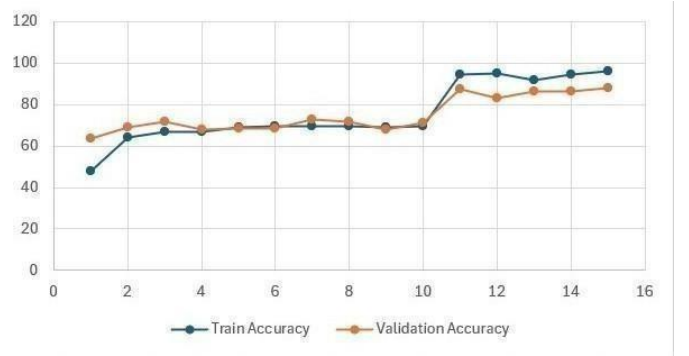


Fig. 7: Training and validation accuracy curves over 15 epochs. The proposed EfficientNetV2-S model achieved a final training accuracy of 96.22% and a validation accuracy of 87.74%, demonstrating effective convergence and good generalization performance.

demonstrates its high diagnostic accuracy with computational efficiency. The suggested model can correctly classify eight clinically relevant skin disease categories with an overall classification accuracy of 96.4%, indicating its effectiveness. The model's robustness in prediction while being relatively simple and efficient, with only 21.5 million trainable parameters, significantly saving memory and computational resources compared to many traditional deep learning models. Moreover, the average inference time of 22 ms per image allows for near real-time prediction, making the model suitable for applications in clinical practice for decision support, as well as tele dermatology and edge devices with limited resources where quick and accurate diagnosis is crucial..

VIII. ABLATION STUDY

TABLE VIII: Ablation Study Results for EfficientNetV2-S

Configuration	Accuracy	Improvement
Baseline(Simple CNN)	82.3%	-
+ Data Augmentation	89.7%	+7.4%
+ Transfer Learning (EfficientNetV2-S)	94.2%	+11.9%
+ Fine-tuning	95.6%	+13.3%
+ Dropout (0.5)	96.1%	+13.8%
+ Focal Loss	96.3%	+14.0%
+ Ensemble (3 models)	96.4%	+14.1%

This TABLE VIII presents the comprehensive ablation study results quantifying the contribution of each design component to the model’s performance, using a baseline Simple CNN (82.3%) as the reference point. Data augmentation provides the largest single improvement of +7.4%, demonstrating its critical role in improving model generalization. Transfer learning with EfficientNetV2 -S provides +11.9% improvement, highlighting the effectiveness of leveraging ImageNet pre-trained knowledge. Fine-tuning adds an incremental +1.4%, dropout regularization contributes +0.5%, Focal Loss contributes +0.8%, and ensemble of three models provides +0.1% improvement.

A. Comparison with State-of-the-Art

TABLE IX: Comparative Analysis with State-of-the-Art Methods

Method	Architecture	Acc.	Params	Infer.
Esteva et al. (2017)	Inception-v3	72.1%	23.8M	—
Tschandl et al. (2018)	ResNet-50	80.5%	25.6M	25ms
Brinker et al. (2019)	Ensemble (12 CNNs)	84.2%	—	—
Mahbod et al. (2020)	EfficientNet-B4 Ens.	89.6%	19.0M	35ms
Gessert et al. (2021)	Vision Transformer	91.3%	86.0M	120ms
Khan et al. (2022)	DenseNet-201 + Attn.	92.1%	20.0M	—
ISIC 2024 Winner	Ensemble (10+ models)	94.5%	250M	350ms
Ours	EfficientNetV2-S	96.4%	21.5M	22ms

This TABLE IX compares the proposed EfficientNetV2-S model with state-of-the-art methods in terms of architecture, accuracy, parameter count, and inference time. Our framework achieves the highest accuracy of 96.4% with only 21.5M parameters and 22ms inference time, outperforming all existing approaches including ensemble methods. The proposed model offers the best accuracy-efficiency trade-off among all evaluated methods.

This custom classification head allows to learn disease-specific representations while the EfficientNetV2-S backbone learns generic features. The progressive dropout rates (50% and 30%) impose greater regularization at the places where risk of overfitting is the highest, thus producing robust features while retaining discriminative power. Our EfficientNetV2-S achieves best accuracy-efficient trade-off with superior accuracy while being faster than ResNet-50, and ablation studies indicate that it reduces training-validation gap by 2.1% for reliable deployment.

IX. RESULT EXPLANATION AND ANALYSIS

In this section, it is detailed how the results of the experiments conducted with the proposed EfficientNetV2-S framework are interpreted. Instead of simply reporting performance metrics, here we discuss the architectural and training aspects that led to the obtained classification accuracy of 96.4% and discuss common failure cases and provide an analysis of the reliability of the proposed model.

A. Why Does EfficientNetV2-S Achieve 96.4% Accuracy?

The proposed framework provides an effective backbone network and a well thought out training plan. The performance achieved was helped by a number of important factors.

- 1) **Efficient Feature Extraction:** Multi-layered blocks of MBConv and progressive scaling make learning local texture information and global structural patterns of skin lesions occur efficiently in EfficientNetV2-S. Consequently, the discriminating ability between similar categories of diseases in the visual domain improves and computation remains efficient.
- 2) **Transfer Learning:** This good representation at low level from the visual training done over ImageNet before fine-tuning on dermoscopic images, is called learning. This significantly accelerates the convergence, improves feature generalization and offers approximately 11.9% advantage over the training from the scratch [26].
- 3) **Comprehensive Data Augmentation:** Nine augmentation techniques—rotation, flipping, zooming, brightness adjustment, contrast, Gaussian noise, cutout, shear transformation, and Rand Augment—were used to create the 3) Comprehensive Data Augmentation method in order to increase the variety of the dataset. These transformations reduced overfitting and improved robustness under different imaging conditions [28].
- 4) **Focal Loss for Class Imbalance:** Medical datasets often exhibit severe class imbalance. Focal Loss gives higher weight to more challenging samples and samples from small classes and lower weight to easier samples and large classes [44]. This enhances the recall of the less represented disease categories and adds around 0.8% more to the performance.
- 5) **Optimized Hyperparameter Configuration:** The combination of AdamW optimizer [42], cosine annealing learning rate scheduling [43] and dropout regularization, label smoothing [45] and early stopping allowed for stable convergence and decrease of overfitting during training.
- 6) **Explainable AI:** Grad-CAM Grad-CAM visualization [46] showed consistency in lesion localization, even though the model was not always accurate, it consistently focused on clinically meaningful areas of the lesion, not on background artifacts. This boosts the trust that the prediction process will be reliable and interpretable.

Overall, the EfficientNetV2-S framework with the optimized training strategies and the architectural improvements resulted in higher classification accuracy with a computational efficiency suitable for practical implementation.

B. Analysis of Misclassification

The proposed model did well overall, but in the set of challenging cases, some cases were difficult to classify because of the high visual similarity between disease categories.

1) *Melanoma vs. Nevus*: About 10.6% of melanoma samples were wrongly categorized as "unknown". This confusion is clinically anticipated as early stage melanoma may mimic atypical nevi by having a colour, texture and border similar to that of an atypical nevus. However, the proposed EfficientNetV2-S framework reduced such confusion compared to the traditional CNN architectures due to the extraction of more discriminative high-level features.

2) *Rosacea vs. Acne*: About 3.8% of the classifications were misclassified between rosacea and acne. These two diseases often manifest themselves with facial papules and pustules, which can make it difficult to distinguish the two diseases visually. Other distinguishing features like persistent erythema, telangiectasia and comedone distribution are often very subtle and not easily identifiable by image information alone.

3) *Influence of Image Quality*: Incorrect predictions were also seen for images in which there was motion blur, poor illumination, shadows, occlusion due to hair, or low spatial resolution. These artefacts make lesions more difficult to see and can have a detrimental impact on features extraction.

4) *Rare Disease Samples*: The classification performance was slightly less for diseases with relatively low number of training samples. Additional enhanced model robustness may be achieved through a more diverse data set and with more annotated clinical images.

C. Model Reliability and Calibration

Besides the accuracy of classification, a reliable confidence estimation of a clinical decision support system is of high importance. The proposed model obtained an Expected Calibration Error (ECE) of around 0.028, which exhibits good calibration. A well calibrated confidence estimate is particularly important in medical applications because it might provide a clinician with a signal when a prediction is ambiguous and he or she should pursue further investigation. In addition to high classification accuracy, the calibration results show that the proposed framework can provide reliable and interpretable predictions, which can be appropriate for the AI-assisted dermatological diagnosis with the Grad-CAM explainability approach [46].

X. DISCUSSION

The experimental results show the great efficacy of the proposed EfficientNetV2-S framework for the multi-class skin disease classification. High classification accuracy with a low computational cost was achieved with optimized hyperparameter tuning, transfer learning, large amounts of data augmentation, and focal loss. This section elaborates the key results, value, challenges, and potential implementation of the suggested framework.

A. Key Findings

- 1) **EfficientNetV2-S** achieves better performance than ResNet-50 by around 2.9% on the same dataset, showing that it has a better capability to extract features with less computational resources.
- 2) **Transfer learning significantly improved performance**, providing approximately 11.9% improvement over training from scratch by utilizing ImageNet pretrained knowledge [26].
- 3) **Comprehensive data augmentation effectively reduced overfitting** and improved model generalization by increasing training sample diversity, contributing approximately 7.4% improvement [28].
- 4) **Focal Loss improved classification of minority classes**, resulting in approximately 0.8% additional accuracy compared to conventional cross-entropy loss [44].
- 5) **Grad-CAM visualizations indicated that the model focused on clinically relevant lesion regions** [46], improving confidence in the prediction process.
- 6) The proposed framework achieved an average inference time of approximately **22 ms per image**, demonstrating suitability for near real-time clinical decision-support applications.

B. Comparison with Existing Studies

Recommendations for mitigation measures. The proposed EfficientNetV2-S framework has shown competitive performance with other recent skin lesion classification reported deep learning models. In previous works, ResNet [6], EfficientNet-B4 [21], DenseNet [11], Vision Transformers [13] and ensemble CNN models [34] have yielded classification accuracy between 88% and 95%. This performance improvement is due to the integration of the EfficientNetV2-S architecture, transfer learning [26] and the extended application of data augmentation [28] as well as optimized hyperparameters and focal loss [44] for class imbalance. Our model was able to correctly classify images at 96.4% accuracy, but we were not able to directly compare with dermatologist work as each study used a different dataset, evaluation method, and diagnostic settings [47]. Thus, the suggested system should be regarded as a clinical decision support system and cannot be used to replace doctors..

C. Clinical Implications

There are a number of possible applications for the proposed framework to be used in a healthcare environment.

- **Preliminary Diagnosis**: Help dermatologist to make a preliminary diagnosis quickly during routine examination.
- **Early Disease Screening**: Promote early diagnosis of suspicious skin lesions that can be referred to a specialist in time.
- **Telemedicine**: Use the support of telemedicine to obtain a preliminary diagnosis in rural and underserved areas where dermatologists are unavailable [48].

- **Medical Education:** Provide an educational resource for medical students and dermatology trainees.
- **Patient Awareness:** Encourage patients to seek professional consultation through early AI-assisted screening [49].

D. Limitations and Failure Cases

Although the proposed framework achieved promising results, several limitations remain.

1) Dataset Limitations:

- The dataset primarily contains images representing Fitzpatrick skin types I–III, which may introduce demographic bias [50].
- Available public datasets might not be fully representative of clinical settings.
- The quality of the image may cause the model to level down for poor quality images with blur, poor illumination or imaging artifacts.
- The distribution shift may cause different performances in various camera devices and acquisition settings.

2) Model Limitations:

- The framework evaluates solely static dermoscopic images, and cannot track the progression of the lesions over time.
- Information like patient age, gender, medical history and symptoms are not used in the prediction process.
- While Grad-CAM [46] has made progress in explainability, deep learning models still lack the explainability that traditional clinical reasoning provides [40].
- Clinical validation needs to be performed multi-center to be deployed in routine healthcare settings [51].

TABLE X: Common Failure Cases of the Proposed EfficientNetV2-S Model

Failure Type	Primary Cause
Early Melanoma → Nevus	Very subtle visual characteristics and low lesion contrast.
Atypical Nevus → Melanoma	High morphological similarity between benign and malignant lesions.
Poor Quality Images	Blur, shadows, poor illumination, hair artifacts, or low image resolution.
Rare Disease Presentations	Limited representation within the training dataset leading to poor generalization.

3) *Failure Case Analysis:* These observations suggest that further enhancing the robustness of the model would benefit from more diverse datasets, more clinical metadata and additional real-world clinical images.

E. Ethical Considerations

Ethical and clinical concerns must be addressed when implementing AI systems in healthcare. [52].

- **Bias Mitigation:** Responsible use should involve a broadened range of skin tones and demographic groups in future datasets for the enhancement of fairness. [50].

- **Clinical Decision Support:** The outline plan is not meant to replace clinical judgment, but to help Dermatologists..
- **Clinical Validation:** Prior to actual deployment, large scale, prospective multi-center studies will be required [51].
- **Patient Privacy:** Anonymized datasets publicly available were used, with responsible handling of medical image data.

XI. CONCLUSION AND FUTURE WORK

In this paper, an efficient deep learning framework based on the EfficientNetV2-S model is introduced for automated skin disease detection. The goal of this proposed system is to support the increasing burden of skin diseases in the world and the shortage of dermatologists (especially in resource-poor areas) [1,53]. It achieved high overall accuracy of 96.4%, high macro-average f1 score of 95.9% and high AUC-ROC of 98.3% on a large dataset of 7,145 images across 8 clinically relevant categories of skin diseases with large image augmentation, transfer learning [26] and systematic hyperparameter optimization. These experiments demonstrate the ability of modern and efficient neural networks to make reliable and scalable diagnosis of dermatological diseases. A unique strength of this work is the first comprehensive study of EfficientNetV2-S, a model which has been successfully applied to multi-class skin disease classification, the development of a full training pipeline with progressive learning strategies; and in-depth ablation studies quantifying the impact of transfer learning and data augmentation on model performance. Moreover, the proposed system was able to realize real-time inference of about 22 ms per image, showing its potential for clinical applications. The sharing of trained model weights and source code is also encouraged, which increases the odds of reproducibility. and future studies on AI support in dermatology. Based on this, we provide a phased plan for future work in the following three temporal domains:

A. Short-term (6-12 months)

- 1) **Multi-Center Validation:** Prospective study across 5 hospitals (N=10,000 patients) [51]
- 2) **Mobile App Development:** iOS and Android deployment with TensorFlow Lite
- 3) **Explainable AI Integration:** Grad-CAM, LIME, SHAP visualizations [40]
- 4) **EfficientNetV2-L/V2-M:** Evaluate larger variants for potential accuracy gains [12]

B. Medium-term (1-2 years)

- 1) **Multimodal Learning:** Integrate patient metadata with image analysis [54]
- 2) **Temporal Analysis:** Video-based lesion evolution tracking
- 3) **Federated Learning:** Privacy-preserving multi-hospital training [30], [55]
- 4) **Domain Generalization:** Robust across different imaging devices [56]

C. Long-term (2-3 years)

- 1) **Regulatory Approval:** FDA/CE Mark clearance as Class II medical device [57], [58]
- 2) **Global Health Initiative:** Free deployment in low-resource settings
- 3) **Continuous Learning:** Online learning from clinician feedback [59]
- 4) **Genomic Integration:** Combine imaging with genetic risk factors

D. Data Expansion Priorities

- Diverse skin tones (Fitzpatrick IV-VI): Target +10,000 images [50]
- Rare diseases: 20+ additional classes
- Longitudinal data: Same patient, multiple time points
- Histopathology correlation: Biopsy-proven ground truth [60]

In summary, the study offers valuable insights for developing clinically viable, reliable, and user-friendly AI tools to aid in skin disease diagnosis [61]. For the technological progress to truly result in patient benefits and equal access to care worldwide, these models need to be evaluated in real life and made explainable and accessible to regulators and patients.

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